

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Underwater Mass Spectrometers for Detection of VOCs and Dissolved Gases

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ABSTRACT

We have developed and deployed mass spectrometry systems that are capable of autonomous underwater operation for chemical analysis. Our primary focus is to analyze coastal marine waters for natural and anthropogenic organic compounds and gases. The high spatial and temporal resolutions of the underwater mass spectrometer (MS) systems has been used to monitor and map chemical distributions in harsh environments. The underwater mass spectrometers are housed in separate pressure vessels that are connected in series for deployment. The front vessel serves as the sampling system and contains a peristaltic pump and a 4 port valve. Sample water and charcoal cleaned sample water are alternately pumped into the hollow fiber membrane in the central vessel by switching the valve. The central pressure vessel contains the MS itself, a membrane probe, the turbo molecular/drag pump and a Card PC. The back vessel contains two dry diaphragm pumps in series that serve as backing pumps for the turbo molecular/drag pump. This approach allows us to evacuate the diaphragm-pump pressure vessel at the beginning of each operation, which extends the endurance of continuous operation to longer than a week. Battery powered deployment duration is currently limited to two hours for the ion trap and four hours for the quadrupole MS. All parts can be activated and controlled via Ethernet from an external computer. For gaseous compounds and very volatile organic compounds, an Inficon Transpector quadrupole residual gas analyzer is used. In various experiments, the analyzer was used to monitor dissolved gasses in shallow marine thermal vents and to monitor volatile organic compounds in wastewater influent, water plant and rivers. For trace compounds and pollutants, a Varian Saturn 2000 ion trap analyzer is used to attain higher sensitivity and the optional advantages of MS/MS systems. The underwater ion trap system has been used to detect plumes of volatile organic compounds in Tampa Bay water. For differing tasks, both underwater MS systems have been used in moored, towed, tethered and autonomous underwater vehicle configurations.